

The empirical recurrence for OEIS A303099, recovered from its tail

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Abstract

OEIS A303099 records “Empirical recurrence of order 90” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 110$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{90} = 77568 \neq 0$, so no further tail satisfies anything shorter; any order-90 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A303099 is “Number of nX5 0..1 arrays with every element equal to 1, 2, 3, 4 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 161, 2913, 58160, 1138331, 22292709, 436394066, 8545589681, ...

The entry states:

Empirical recurrence of order 90 (see link above)

As of the “Last modified” line on the live entry (Apr 18 2018 12:36:01, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 110$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 110$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 90: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 90.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 220$ exact terms from $n = 2$ on, it returns order 90 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	15	c_{24}	−202830273593	c_{47}	−122601338921910	c_{70}	−271401919995
c_2	127	c_{25}	−1146573878573	c_{48}	37569942422877	c_{71}	1802559330531
c_3	−352	c_{26}	38808822216	c_{49}	−70450615090459	c_{72}	−360161962281
c_4	−7403	c_{27}	−610903789551	c_{50}	−21344461572836	c_{73}	−482291151781
c_5	−10351	c_{28}	808761495447	c_{51}	−139399264237414	c_{74}	−313989427380
c_6	172454	c_{29}	2766072678761	c_{52}	−41386754476563	c_{75}	−45022889508
c_7	655184	c_{30}	−3848481643781	c_{53}	65989748791158	c_{76}	187661419028
c_8	−1559493	c_{31}	1320417964110	c_{54}	72710603275758	c_{77}	5621659018
c_9	−13149585	c_{32}	1961816539907	c_{55}	28953207268643	c_{78}	23274977574
c_{10}	−2972842	c_{33}	10565417058080	c_{56}	66599328492008	c_{79}	38360903044
c_{11}	123596467	c_{34}	38960399768925	c_{57}	49008497861013	c_{80}	−13469288732
c_{12}	192241517	c_{35}	2169261707348	c_{58}	111957653192629	c_{81}	−3877784196
c_{13}	−670101555	c_{36}	−19909486788239	c_{59}	66320489200980	c_{82}	−1826640312
c_{14}	−1438057719	c_{37}	−13115944520243	c_{60}	61729672405411	c_{83}	293023864
c_{15}	2577124452	c_{38}	−36153016074511	c_{61}	27261994094496	c_{84}	264674544
c_{16}	5450093418	c_{39}	−48304743835531	c_{62}	54719919526885	c_{85}	148636688
c_{17}	−10287967814	c_{40}	−72711902589220	c_{63}	22027722134456	c_{86}	27668464
c_{18}	−18433764024	c_{41}	−117069684513825	c_{64}	9179705896415	c_{87}	−8574176
c_{19}	34994482303	c_{42}	−65674158389344	c_{65}	−316102096411	c_{88}	−2303488
c_{20}	52757107653	c_{43}	65122810905635	c_{66}	−587919163288	c_{89}	232064
c_{21}	−99062425061	c_{44}	50801082379788	c_{67}	6929111600982	c_{90}	77568
c_{22}	−24462388905	c_{45}	145534620149129	c_{68}	1520502070668		
c_{23}	457482953691	c_{46}	−39425569941603	c_{69}	−1157497062592		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A303099 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{90} - c_1x^{90-1} - \dots - c_{90}$. Its constant term is $-c_{90} = -77568$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 90 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 90, so $\deg q = 90$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 90, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 90, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 77568.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-90 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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